

Problem-Solution Fit

Date	08-07-2026
Team ID	
Project Name	Automated Credit Card Approval Prediction using Machine Learning
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Primary: Individuals applying for credit cards through banks or financial institutions. Secondary: Bank credit officers and loan verification staff. Tertiary: Bank managers and financial institutions monitoring customer risk.	2. PROBLEMS / PAINS + ITS FREQUENCY [PR] P1: Slow manual credit card approval process — Every application. P2: Incorrect approval/rejection decisions due to human error — Frequently. P3: Difficulty assessing applicant eligibility accurately — Daily. P4: Large number of applications increases processing time — Every working day.	3. TRIGGERS TO ACT [TR] T1: Customer submits a credit card application. T2: Bank receives a high volume of applications. T3: Need for faster and more accurate approval decisions. T4: Requirement to reduce financial risk and fraud.
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4. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL]

Budget: Customers prefer a free online application process.

Devices: Applications are submitted through smartphones, laptops, or desktops.

Knowledge: Many applicants are unaware of bank eligibility criteria.

Internet: Stable internet connection is required for online application.

6. BEHAVIOR + ITS INTENSITY [BE]

High: Applicants frequently check their application status.

Medium: Customers compare credit card offers from different banks.
High: Bank officers verify applicant documents before approval.

Result: Faster decisions improve customer satisfaction and reduce workload.

7. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Bank Website
- Mobile Banking Application
- Internet Banking Portal
- Email & SMS Notifications

OFFLINE

- Bank Branch
- Customer Service Desk
- Financial Advisors

5. AVAILABLE SOLUTIONS PROS & CONS [AS]

Pros: Traditional bank verification systems are secure and trusted.

Cons: Manual verification is time-consuming and prone to human error.

Pros: Online banking portals provide digital applications.

Cons: Most systems lack intelligent Machine Learning-based prediction.

8. PROBLEM ROOT / CAUSE [RC]

R1: Manual verification consumes significant time.

R2: Human decisions may be inconsistent.

R3: Large application volumes delay processing.

R4: Lack of intelligent prediction increases operational workload.

9. YOUR SOLUTION [SL]

Credit Card Approval Prediction System

- Applicant enters personal and financial details through a web application.
- Machine Learning model analyzes applicant information such as age, income, employment status, credit history, loan amount, and existing debt.
- The trained model predicts whether the application is **Approved** or **Rejected**.
- Flask-based web application provides instant prediction results through a simple and user-friendly interface.
- The application is deployed online using **Render** and maintained using **GitHub**, making it accessible from any modern web browser.

